

Deep learning-based vehicular engine health monitoring system utilising a hybrid convolutional neural network/bidirectional gated recurrent unit

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ABSTRACT

Vehicles play a pivotal role in the current era of Industry 4.0 by providing passengers with excellent mobility, comfort, and safety while strengthening national and international economies. Unanticipated vehicular engine issues can hinder performance and lead to costly maintenance. As analytics processes become faster, more accurate, and more reliable, intelligent fault prediction and diagnosis for vehicles, particularly engines, is becoming increasingly popular. To date, hybrid deep learning approaches to vehicle engine diagnostics have been limited, and none have used engine health monitoring and categorisation based on vulnerability assessment and vehicle structural information. This paper introduces a hybrid deep learning-based vehicular engine health monitoring system (VEHMS) decision model using Deep CNN (convolutional neural network)-BiGRU (bi-directional gated recurrent unit). This model monitors a vehicle's engine health in real-time and classifies its status as good, critical, moderate, or minor condition. Several advanced and hybrid deep learning algorithms were applied to monitor engine health and categorise its status by integrating sensor data with evaluated vulnerability information from an infrastructure vulnerability assessment model. The Deep CNN-BiGRU-based VEHMS decision model outperformed other techniques with an accuracy of 0.8897, ensuring minimal decision losses while classifying engine conditions. This study aims to contribute to developing comprehensive vehicle health monitoring systems and advance the automotive industry by incorporating more intelligent features. The proposed approach can enhance vehicle performance, reliability, and efficiency in the transportation sector by improving engine health monitoring.

1. Introduction

The vehicle acts as a key factor in our society for prosperity, employment, economic exchange, and personal freedom (Göhlich, Syré, van der Schoor, Jefferies, Grahle, & Heide, 2022). The engine is the heart of the vehicle, which is made up of a variety of vital components and systems. Ensuring the desired efficiency and maintaining the performance and durability of the vehicular engine is the most challenging task. Various studies about real-time vehicle engine monitoring have been carried out over several decades to offset such challenges, including engine oil level, engine temperature (Thamaraimanalan, Mohankumar, Dhanasekaran, & Anandakumar, 2021), and engine exhaust (Sumathy et al., 2023) monitoring systems to predict

its health condition. The current vehicles are equipped with OBD-II to monitor the vehicle's engine health and ensure a comfortable journey with features like navigation systems, anti-theft systems, lane detection systems, collision mitigation systems, and driving assistance systems. Typically, OBD-II is used to monitor specific characteristics of vehicle engines, such as the rotational speed, coolant temperature, exhaust gas quality, air-fuel ratio, and ignition system, in addition to these primary concerns relating to the vehicle's structure (Shafi, Safi, Shahid, Ziauddin, & Saleem, 2018). As the importance of automobiles in our everyday lives is increasing, the need to monitor the condition of a vehicle's engine in a more sophisticated manner is also becoming more crucial.

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On the other hand, engines have many moving parts that can degrade under extreme operating conditions due to the lack of timely maintenance initiatives. Additionally, in certain circumstances, maintenance operations depend entirely on expert judgment based on noise observations, which can be misleading and lead to unexpected maintenance expenses without a satisfactory resolution. Failure to recognise minor abnormalities may also result in severe damage to the entire engine subsystem, leading to significantly higher maintenance costs for the owner. Users often rely on past experiences and technical opinions to identify vehicle faults by listening to the engine's sound, although this method is not always successful in certain instances. Moreover, this action can also cause significant degradation of the vehicle's engine and damage its essential components. As a result, various relevant studies have been conducted lately to develop robust, fast, reliable, and accurate monitoring systems for assessing vehicle engine health using modern techniques and instruments (Madain, Al-Mosaiden, & Al-khassawneh, 2010). In this case, data-driven techniques are ideal for fault prediction because they are more appropriate and dependable for diagnosing engine faults than knowledge-based systems that use artificial intelligence (AI) (Fan, Hsu, Tsai, He, & Cheng, 2020; Hou, Lee, Lv, & Keung, 2023). Various methods and techniques are available and extensively employed in the automotive sector to detect and notify information about the raised inconsistency in vehicle engines.

Currently, the available monitoring functions of vehicular engines are of little utility to both the vehicle and the user. Furthermore, they are incompatible with the Industry 4.0 era since user time management is required to increase productivity and ensure a comfortable and enjoyable journey. Because these facilities are unable to predict vehicular conditions, users are forced to deal with unexpected difficulties, unwanted vehicular failures, inconsistent performance, engine health degradation, and unpredictable maintenance requirements. Therefore, users are unable to utilise their valuable time effectively. Recently, various AI methods, such as machine learning (Shafi et al., 2018), deep learning (Al-Zeyadi, Andreu-Perez, Hagrass, Royce, Smith, Rzonowski, & Malik, 2020; Rahim et al., 2022), fuzzy logic (Rahim et al., 2021), and artificial neural networks (ANN) (Cervantes-Bobadilla, García-Morales, Saavedra-Benítez, Hernández-Pérez, Adam-Medina, Guerrero-Ramírez, & Escobar-Jiménez, 2023), have been used to diagnose vehicular engine issues in the best possible way. Although, the deep learning-based technique has shown promise in the automotive industry, with reasonable accuracy and performance results when dealing with large amounts of data (Rahim et al., 2022); unfortunately, its applicability is limited and engines continuously produce tons of complex data. On the other hand, it is being used prominently for different applications in a wide range of industries to handle massive complex data, including image processing (Khandokar, Hasan, Ernawan, Islam, & Kabir, 2021; Lambert, Forbes, Doyle, Dehaene, & Dojat, 2024), video processing (Cob-Parro, Losada-Gutiérrez, Marrón-Romera, Gardel-Vicente, & Bravo-Muñoz, 2024; Islam, Sultana, Kumar Roy & Al Mahmud, 2020), speech processing (Fan, Yin, Yu, & Gilliland, 2020; Sheng et al., 2024), and text processing (Islam, Ab Ghani, & Ahmed, 2020; Islam & Ghani, 2021; Khanday, Rabani, Khan, Rouf, & Mohi Ud Din, 2020; Khandokar et al., 2021), with satisfying results. Additionally, hybridisation AI by combining different deep learning techniques facilitates achieving results faster with higher accuracy. This is particularly beneficial for processing inconsistent and uneven data, as individual techniques may not be suitable. For example, a combined convolutional neural network (CNN) and bi-directional gated recurrent unit (BiGRU) could extract deep characteristics of charging data and identify faults promptly to provide early warning and assist in avoiding severe accidents during charging (Gao, Du, Zhang, & Yang, 2024). Another hybrid AI was proposed in (Ertargin, Yildirim, and Orhan (2024) by combining long short-term memory (LSTM) with CNN to monitor and prognose electrical and mechanical faults of induction motors. So, hybrid AI shows hope to process massive amounts of complex vehicular engine data and assess its health condition accurately.

Another noteworthy point is that prior research studies in this field have used data from the OBD-II (Shafi et al., 2018). In some instances, research has been conducted on individual components of vehicle engines (Du et al., 2021) rather than taking into account any risk considerations and the vulnerability problem of each component. Furthermore, we have not found an advanced vehicle health monitoring system that incorporates consideration of risk factors influencing engine performance or prioritises maintenance urgency based on the engine's health conditions. As a result, we intended to develop an advanced vehicular engine monitoring system that can accurately and efficiently classify the engine's health condition while employing structural information for preventative maintenance. In this case, a hybrid deep learning-based vehicular engine condition prediction decision model is proposed to overcome these limits. The proposed vehicular engine health monitoring system (VEHMS) model can determine the importance of engine maintenance based on the state of the engine health, i.e., critical, moderate, and minor, utilising structural information of vulnerable components. Although this study makes a significant contribution, the most important contributions are as follows:

1. Develop a new hybrid deep learning algorithm for making faster and more accurate decisions regarding the health of a vehicle's engine.
2. Evaluate the findings with other techniques to come up with a recommendation for the best vehicle engine health monitor for the automotive sector.
3. Create an ideal model to bring the whole vehicular system under an advanced and intelligent surveillance system for the future.

Now, this research planned to employ Deep CNN-BiGRU as the hybrid deep learning approach to develop the Deep CNN-BiGRU-based VEHMS decision model to monitor the vehicular engines' health condition in real-time. In this case, we fed the data to this algorithm as a 0.8:0.1:0.1 ratio for training, validation, and testing purposes, respectively. We found that this approach provided satisfactory decision accuracy with 0.8897, besides excellent receiver operating characteristic (ROC) curve and minimum decision losses.

The remaining article presents an extensive literature study to acquire knowledge on real-time vehicular engine conditions monitoring systems and identify probable solutions through profound research on developed decision models as the theoretical background study in Sections 2 and 3, respectively. After that, in Section 4, we discussed the research methodology to achieve the decision accuracy of the decision model. Then, Section 5 evaluates the proposed model's performance and analyses it with other approaches to ensure the best outcomes. Finally, this research concludes with future directions to enhance and extend the VEHMS decision model as an advanced system to move the automotive sector forward with compatibility of Industry 4.0 by Section 6.

2. Related research work

Recently, AI techniques have been used prominently to monitor mechanical system conditions and diagnose faults in various sectors. Especially in the automotive sector, several faults plague vehicular systems due to many parts with their functionalities. As modern automotive systems have been improved with massive features and advanced technology like self-driving, malfunctions have increased substantially from the turn of the 20th century to the present day. GPS, biometric access to the vehicles, air-conditioning system, security system, etc. The fault diagnosis of vehicular systems is an essential requirement that increases the vehicle's reliability by monitoring the state of sensible parts. Much research has been conducted to understand the problem by diagnosing faults from various perspectives (Al-Zeyadi et al., 2020). More specifically, in recent times, AI-based diagnosis techniques based on various advanced approaches have been increasingly used in

vehicles to diagnose faults due to their significant effects (Kumar & Daudpoto, 2019). The authors of Jeong, Son, Jeong, and Lee (2018) proposed an AI-based self-diagnosis system for autonomous vehicles to speed the process of self-diagnosis while simultaneously decreasing administrative burden. The first of the three modules in the proposed system gathers information from autonomous vehicles through the Internet of Things. Then, a deep learning optimised decision model uses the information collected in the first and second modules to develop a training dataset. Finally, the third component is an AI-based edge computing self-diagnostic model which speeds up diagnosis processes and minimises the system overhead. Again, González et al. proposed an AI-based decision-making system for fault identification using an auto-associative neural network (AANN) and support vector machine to classify diagnosed vehicular faults (González & Pablo, 2018). Sun, Yu, Chen, and Kadam (2019) proposed an AI-based predictive decision model to detect the anomaly with time series to reduce vehicular downtime using a Generative Adversarial Network (GAN). In this case, the threshold is generated through prediction using deep learning/machine learning algorithms.

Similarly, Zhang et al. proposed a back propagation neural network AI system to diagnose electric vehicle battery faults (Zhang et al., 2024). Besides, another intelligent decision model was proposed to diagnose electric vehicle motor and inverter faults using the machine learning approach (Murphey, Masrur, Chen, & Zhang, 2006). A novel fault diagnosis and classification decision model was developed using a similar vehicular power transmission system approach. In this case, machine learning approaches achieved higher accuracy and sufficient effectiveness than other approaches (Gong, Su, & Tseng, 2020). Meckel's research team utilised the same approach to develop an AI-based fault diagnosis model for hybrid-electric vehicle driver assistance and safety systems. They also found that this approach performs efficiently in diagnosing faults in hybrid-electric vehicles (Meckel, Schuessler, Jaisawal, Yang, & Obermaier, 2020). Moreover, every day, the fault diagnosis model's advancement is increasing to ensure the prediction model's accuracy using hybrid deep learning and deep learning approaches. In continuation of that, Yin et al. proposed a fault diagnosis decision model for the vehicular on-board equipment of high-speed railways using KNN (k-nearest neighbour), DBN (deep belief network), and ANN-BP (artificial neural network with back propagations) approaches (Yin & Zhao, 2016). Guo, Zhong, Ji, Liu, and Yang (2018) employed a convolutional neural network (CNN) to monitor the installed sensors in the unmanned aerial vehicle (UAV), where they utilised the time–frequency map to diagnose UAV sensor faults through their experiments. The same approach for intelligent vehicles showed satisfactory outcomes by learning autonomously from extracted features and data during the voltage anomalies among the designated components (Haque, Shaheed, & Choi, 2018). Besides these approaches, as a hybrid AI, both long short-term memory (LSTM) and bidirectional long-term memory (Bi LSTM) are also used prominently for identifying the faults of different types of vehicles (Sun et al., 2023). For example, (Sadhu, Anjum, & Pompili, 2023) proposed an anomaly detection architecture in UAV by deep autoencoder network based on CNN and Bi LSTM. Also, they enhanced this proposition by considering the classification matter about detected anomalies to an internal fault intrusion attack or crash possibilities using a combined CNN and LSTM-based classifier.

On the other hand, the implementation of GRU and Bi-GRU for fault detection of vehicles to mitigate arisen complexities is also evident due to satisfactory outcomes (Feng et al., 2023). Khaliq proposed the vehicle fault detection system with automatic feature learning capabilities as part of the health monitoring system by minimising labour extensiveness compared to conventional approaches (Khaliqi & Iulian, 2023). Again, Xia's research team recently employed a multi-layer perceptron (MLP) approach with an attention mechanism to sort the raised faults by capturing the semantic relationship between monitoring variables and faults in the autonomous underwater vehicle. In this case, they

utilised Bi-GRU to ensure automatic feature engineering to gain monitoring variables' time-based dynamic characteristics (Xia, Zhou, Shi, Li, & Xu, 2021). Similarly, hybrid deep learning is performed more outstandingly in some cases than popular deep learning approaches. Such promising research work encourages the efficient use of this approach for condition monitoring and fault identification of mechanical systems. For example, Shao's research team obtained more than 0.96 fault detection accuracy using a CNN-LSTM-based approach for monitoring electric vehicle charging piles, where CNN automatically extracts the features outstandingly (Shao & Kim, 2018, 2024). Again, for gear fault diagnostic purposes, this approach also represented satisfactory performances to reduce downtimes, improve operation reliability, and minimise operations and maintenance expenses (Yang & Wang, 2023). Similarly, the CNN-LSTM approach showed nearly 0.93 accuracy, and in this case, CNN was responsible for extracting features, and LSTM was responsible for exploring internal connections and identifying the faults of the dedicated equipment in vehicles (Guo et al., 2023). In various cases, such deep learning and hybrid deep learning approaches were employed and achieved satisfactory outcomes (Shaikh, Hussain, & Chowdhry, 2023). The above literature study shows that these approaches are not yet employed in the vehicular engine to diagnose the vulnerable components and categorise the engine condition for maintenance priority as critical, moderate, minor, or good condition. Furthermore, the recent research advancements in health monitoring systems for vehicles and their engines are presented in Table 1.

From the above literature, it is evident that there is a good prospect for designing vehicle or engine health monitoring systems using advanced AI techniques, such as hybrid deep learning, to achieve more accurate and rapid decision-making. Additionally, novel concepts like the introduction of vulnerability assessment techniques with the engine's structural real-time information will create a new dimension to set maintenance priority classification based on engine health conditions. Therefore, to develop an advanced vehicular engine condition monitoring and diagnosis system based on real-time data, this work plans to employ CNN, LSTM, BiLSTM, GRU, BiGRU, CNN-LSTM, Attention CNN-LSTM, and Deep CNN-BiGRU approaches. The notable matter is that we have incorporated the vulnerability assessment technique and engine structural information in this work to categorise the maintenance priority of the vehicular engine and effectively minimise future maintenance expenses.

3. Theoretical background

3.1. VEHMS decision model

From Rahim et al. (2022), the severity value to measure the maintenance priority for δ^{th} vulnerable component (VC)/ vulnerable point (VP) of the vehicular engine is

$$SV_{\delta}(t) = (1 - \lambda)^k * S_{\delta} * W_{\delta} * \left\{ \sum_{\zeta=1}^m I_{\zeta} \times IRI_{\zeta} \right\} \quad (1)$$

where δ represents the number of VC/VP as $\delta = 1, 2, 3, \dots, n^{th}$ subsystem in the engine and ζ signifies the number of intensifying factors of δ^{th} VC/VP as $\zeta = 1, 2, 3, \dots, m^{th}$. S_{δ} is attained based on the sensor reading and W_{δ} is the weightage of δ^{th} VC/VP. The weightage W_{δ} can be defined as the ratio between the significance of δ^{th} VC/VP to the significance of all VC/VP, where $\sum_{\delta=1}^n W_{\delta} = 1$. Additionally, I_{ζ} and IRI_{ζ} is **individual value** and **intensity of relative importance** of ζ^{th} intensifying factor of δ^{th} VC/VP. I_{ζ} could be obtained by following the **vulnerability assessment questionnaire technique** (Table 4) (Rovito, 2016) with the **common vulnerability scoring system (CVSS) qualitative severity evaluation scale** (rating and indexing scale) (Holm & Afridi, 2015) as impact severity on engine performances. Here, the rating and indexing scale for impact severity are 0.0 for no impact, $0 < \text{low impact} \leq 3.99$ for, $3.99 < \text{medium impact} \leq 6.99$, $6.99 < \text{high impact} \leq 8.99$, and $8.99 < \text{critical}$

Table 1

Recent research advancements about vehicular health monitoring system.

References	Implemented technique	Research scope & advantages	Drawbacks & challenges
Vasavi, Aswarth, Pavan, and Gokhale (2021)	Ensemble of Artificial neural network and K-nearest neighbours	Real-time vehicular health is monitored based on data from OBD-II.	Due to using only data from OBD-II, the implemented technique will not be able to detect faults in the crankshaft and cylinder-piston. Additionally, the proposed system is unable to prioritise maintenance for any specific components.
Gomes and Wolf (2021)	Dynamic Bayesian Network	Monitoring system for autonomous vehicles includes a dynamic control module, electronic control units, communication, and localisation system	This research was performed using the sensor data only instead of considering environmental factors that may degrade vehicular health. Also, the research team avoided incorporating vulnerable assessment techniques to ensure road user's safety and reliability.
Rahim et al. (2022)	Stacked ensemble deep learning	Real-time vehicular engine health monitoring system considering environmental influential factors and system vulnerability.	In this research, the performance accuracy of the proposed system is comparatively lower and bounded within 80.3%. Also, avoid implementing hybrid deep learning techniques to analyse the proposed model's performances.
Shafi et al. (2018)	Support Vector Machine	Remotely vehicular health monitoring system using OBD data.	The research team proposed the VHMS by using only OBD data about the exhaust, cooling, and ignition systems. They only implement conventional AI techniques instead of implementing prominent and effective approaches like the hybrid deep Learning technique. Also, the developed system could not categorise the vehicle's health condition into critical, moderate, minor or good condition for maintenance priority. It just follows a conventional process based on a predefined level.
Das, Kumar, and Krishna (2024)	K-nearest neighbours and decision tree technique	Monitoring the electric vehicular battery health	The research work is only confined to monitoring battery health instead other significant components of the vehicle. They also could implement hybrid deep learning techniques like Deep CNN-BiGRU to analyse the battery health which is overlooked.

impact ≥ 10 . IRI_{ζ} is the ratio between the relative importance of an intensifying factor to the total relative importance of all intensifying factors of individual VC/VP, where $\sum_{\zeta=1}^m IRI_{\zeta} = 1$. Both IRI_{ζ} and W_{δ} could be calculated by employing **value focused thinking (VFT)** method with 0 to 10 scale which are described in Section 4.1.1 and presented in Tables 5 and 6, respectively. The notable matter from Eq. (1), $\left\{ \sum_{\zeta=1}^m I_{\zeta} \times IRI_{\zeta} \right\}$ is termed as **vulnerability affecting value** and $W_{\delta} * \left\{ \sum_{\zeta=1}^m I_{\zeta} \times IRI_{\zeta} \right\}$ is the VC/VP's **vulnerability value**. Also, λ is the degradation of VC/VP's performances for k_{th} 10,000 km vehicle movement. Now, based on the severity value in Eq. (1) the following threshold ($TH_C(745.3) > TH_M(407.79) > TH_{MN}(99.0)$) could be utilised to categorise δ^{th} VC/VP's health condition as

$$\begin{cases} \text{Critical } (SV_{\delta C}), & \text{if } SV_{\delta}(t) \geq TH_C \\ \text{Moderate } (SV_{\delta M}), & \text{if } TH_C < SV_{\delta}(t) \geq TH_M \\ \text{Minor } (SV_{\delta MN}), & \text{if } TH_M < SV_{\delta}(t) \geq TH_{MN} \\ \text{Good } (SV_{\delta G}), & \text{if } SV_{\delta}(t) < TH_{MN} \end{cases}$$

In this case, these values are assumed based on the minimum severity values of the most prominent VC/VP in the engine system. Since the above-stated boundary limits define the individual VC/VP health conditions in the engine subsystem. In this context, for t time interval, the following (number of VC/VP $\times$ number of VC/VP health condition or $n \times 4$) matrix (red boundary in Fig. 1) is prepared to decide the whole engine's health condition in real-time. A vehicular engine is indeed composed of many components. In this case, it is very tough to categorise based on priority for maintenance if one more primary VC/VP exhibits inconsistent behaviour at the same time. Also, if one more secondary or lower secondary VC/VPs exhibits inconsistent behaviours and has the same severity value. Then it is hard to categorise them for maintenance based on the maximum severity value. Also, based on the maximum value in this context the system may notify us that the vehicular engine has a minor problem. Therefore, aggregating all elements column-wise of this decision matrix (Fig. 1), the VEHMS decision model for the complete vehicular engine at t time with decision-making

threshold ($TH_{DC}(925.29) > TH_{DM}(478.79) > TH_{DMN}(99.0)$) is

$$\sum VEHMS_D(t) = \sum SV_{\delta C} + \sum SV_{\delta M} + \sum SV_{\delta MN} + \sum SV_{\delta G} \quad (2)$$

where $\delta = 1, 2, 3, \dots, n$ and engine's condition in real-time is

$$\begin{cases} \text{Critical,} & \text{if } \sum VEHMS_D(t) \geq TH_{DC} \\ \text{Moderate,} & \text{if } TH_{DC} < \sum VEHMS_D(t) \geq TH_{DM} \\ \text{Minor,} & \text{if } TH_{DM} < \sum VEHMS_D(t) \geq TH_{DMN} \\ \text{Good,} & \text{if } \sum VEHMS_D(t) < TH_{DMN} \end{cases}$$

By utilising the stated thresholds, this decision model can decide whether the engine's health is in good condition or has a minor, moderate, or critical problem. Notably, these thresholds are assumed based on all VC/VPs functionalities, damage possibilities, and potential to affect other VC/VPs due to inconsistencies with consolidated minimum severity values.

3.2. Hybridisation using BiGRU and CNN

Two different types of neural networks, Deep CNN and RNN BiGRU, are planned for use in Deep CNN-BiGRU-based vehicle health monitoring systems. The Deep CNN-BiGRU model contains specific layers termed BiGRU, Deep CNN and hierarchical aggregation layer during the training phase. In this case, we have embedded the layer with tokenisation at the beginning of our model. We also ensured that the BiGRU has a one-dimensional input sequence with spatial dropout for extracting dependent features with long distances. It then moved its output to the dynamically managed Deep CNN layer, allowing the vector output to be hierarchically aggregated from the preceding CNN output. Finally, hierarchical aggregation layer output is linked with categorical entropy loss function with ADAM optimiser to categorise the engine condition into multiclass (four classes, i.e., critical, moderate, minor, and good). The hybridisation using BiGRU and CNN are described as follows:

VC\VP ↓	Condition →	Critical ($SV_{\delta C}$)	Moderate ($SV_{\delta M}$)	Minor ($SV_{\delta MN}$)	Good ($SV_{\delta G}$)
V_1		SV_{1C}	SV_{1M}	SV_{1MN}	SV_{1G}
V_2		SV_{2C}	SV_{2M}	SV_{2MN}	SV_{2G}
V_3		SV_{3C}	SV_{3M}	SV_{3MN}	SV_{3G}
V_4		SV_{4C}	SV_{4M}	SV_{4MN}	SV_{4G}
V_5		SV_{5C}	SV_{5M}	SV_{5MN}	SV_{5G}
-		-	-	-	-
-		-	-	-	-
V_n		SV_{nC}	SV_{nM}	SV_{nMN}	SV_{nG}

Fig. 1. $n \times 4$ Decision Matrix.

3.2.1. BiGRU layer

This layer consists of BiGRU simultaneously sequence of action. With specific calculations, GRU cell processing can be seen below. For the resetting gate and the concealed state, the input x is used in this instance.

$$r_t = \sigma(W_{xr}X_t + W_{hr}h_{(t-1)} + b_r) \quad (3)$$

$$z_t = \sigma(W_{xz}X_t + W_{hz}h_{(t-1)} + b_z) \quad (4)$$

$$\hat{h}_t = \tan h(W_{xh}X_t + W_{hh}(r_t \odot h_{(t-1)}) + b_h) \quad (5)$$

$$\hat{h}_t = z_t \odot h_{(t-1)} + (1 - z_t) \odot \hat{h}_t \quad (6)$$

In this case, W indicates matrices, b used for parameters, $\odot$ for elements element-wise multiplication. Here, i_t is used for the input gate, o_t do the task of the output gate, and f_t is for forget gate. σ used for activation function. W_x is used for weights of different gates(x). $h_{(t-1)}$ is the output from BiGRU with timestamp $t - 1$, X_t is the current timestamp.

BiGRU is especially beneficial when the context for the feedback is critical. It is a term that is widely used to describe class prediction. With bi-directional GRU, data does not merely go back and forth in two hidden nations. As a result, the Bi-GRUs are much more familiar with the issue. BiGRUs are used to scale up the network-useable input information pieces. BiGRU sequences are as follows:

$$\overline{bigru}_t = \overline{BiGRU} * (\bar{h}_{(t-1)}, X_t) \quad (7)$$

$$\overline{bigru}_t = \overline{BiGRU} * (\bar{h}_{(t-1)}, X_t) \quad (8)$$

$$bigru_t = (\bar{h}_t, \bar{h}_t) \quad (9)$$

As a result, the BiGRU encoder's output is a series of vectors as follows:

$$BG = [bg_1, bg_2, \dots, bg_N] \in R^{(N \times d')} \quad (10)$$

3.2.2. Deep CNN layer

It is one of our model's fundamental layers. This extensively enlarged convolution layer aims to recover hierarchy multi-granularity characteristics as a text-based consent form. Unlike traditional CNNs, which perform convolutional layers directly on the training set embeddings, our Deep CNN uses Bi-GRU output Vectors, which incorporate contextual information. For the first block of matrix multiplication, we state it as follows.

$$UHV^0 = BG = [bg_1, bg_2, \dots, bg_N] \in R^{(N \times d')} \quad (11)$$

The original input variable d is contained in the input matrix. As a result, the results within each intermediate structure might be considered at the end block.

$$UHV^1 = [uhv_1^l, \dots, uhv_N^l] \in R^{(N \times k)} (l \in [1, L]) \quad (12)$$

where L is the total amount of blocks and k denotes the number of filters per block. Now take the l th number blocks, let

$$W^l \in R^{(k \times w \times k)} W^1 \in R^{(k \times 1 \times d')} \quad (13)$$

This layer matrix now uses fully convolutional w input vector data from the k filter kernel. The following is the order in which two close blocks must be transitioned.

$$UHV = f(W^l, UHV^{(l-1)}) \quad (14)$$

Selector filters are utilised as a viewport over the w -length entry since f is used as a tuned function. Usually, $uhv_t^l \in UHV^l$ is presented as

$$uhv_t^l = \text{ReLU}(W^l \oplus [uhv_{(t+1)r}^{(l-1)}]_{(t=0)}^{(w-1)}) \quad (15)$$

where $\oplus$ is a concatenation operator of convolution, r , for deep rate with CNN. Rectified linear units (ReLU) are the activation function. We utilise a deep mechanism of the Deep CNN layer. The dilation rate is twice at each block with a zenith rate $2^{(L-2)}$ and each field of block is extended by maximum width $(w - 1)2^{(L-1)}$. Now, $r = 1$ for standard dilation (Yu & Koltun, 2015). Finally, we get hierarchical maps of $UHV^1, UHV^2, \dots, UHV^L$.

3.2.3. Aggregation layer

To capture and preserve dense Hierarchical representations, we may use 1D and Deep CNN convolution, which corresponds to different kernel dimensions, and this layer serves as the aggregate. Dropout is used in the multi-cycle to make the models unfavourable for overfitting (Yu & Koltun, 2015).

For the aggregate layer 1D convolutional and transition matrix, W_j is used. Weight W_{ij} codes are essential for spatial interactions between low-level and higher-level characteristics. The designated point is the most notable feature of this 3D curved vector. The scale used by CNN is denoted by d , while the dilated CNN number is denoted by n .

Can the result be calculated by the size $(1, i[-1], n * d)$. Let u be $(none, s, i[-1])$ specified, then $(uhv_{(j|j)}) = \text{conv1d}(uhv, W)$. This vector result is passed to the hierarchical aggregation layer. The outcome feature of one convolutional block considers, for example, and formally arrives as the deep convolutional procedure in detail. Now $UHV^1 = [uhv_1^l, \dots, uhv_N^l] \in R^{(N \times k)} (l \in [1, L])$ indicates the result of l th convolutional block.

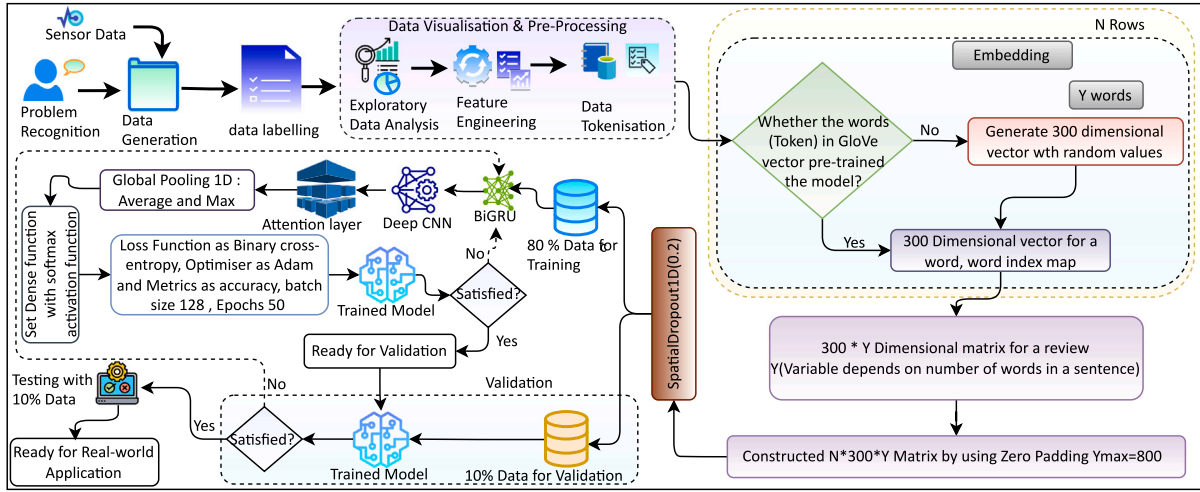


Fig. 2. Flowchart of the DEEP CNN-BiGRU Approach.

Let us use uhv_i in the input aggregation layer, as well as the prediction vector ($uhv_{(j|i)}$) for viewing raw features to transfer, which is obtained by multiplying uhv_i by the linear transformation W_j .

$$(uhv_{(j|i)}) = uhv_i * W_j \quad (16)$$

b_{ij} is set to zero in the “softmax routing” function and changes with the a_{ij} scale agreements. Its agreement a_{ij} is calculated using the following scale.

$$a_{ij} = (uhv_{(j|i)}) \quad (17)$$

$$b_{ij} = b_{ij} + a_{ij} \quad (18)$$

The primary goal of this layer is to generate a specified and aggregate concept vector by deciding on using each CNN output as just an input. At the focus layer, the following process takes place.

$$a_i = q^T * uhv_i \quad (19)$$

The q , in this case, is a trainable pattern vector. The weighted sum is then calculated and transmitted to the application categorisation overall goal lengths, resulting in a set long aggregation vector.

$$o = \sum_i a_i * uhv_i \quad (20)$$

3.2.4. Prediction layer

The primary goal of this layer is to determine the probability distribution with $p(y|S)$, where y is the class target. Fixed-length and aggregation-based vector o are provided by the softmax function to the Multi-Layer Perceptron classifier:

$$p(y|S) = p(y|o) = softmax(MLP(o)) \quad (21)$$

4. Methodology of vehicular engine health monitoring decision process

This section demonstrates the step-by-step methods to analyse the vehicular engine health monitoring decision model. Before that, it describes the procedures of data generation, data pre-processing, and hybrid deep learning model development eloquently, which is represented by Fig. 2.

4.1. Data generation

4.1.1. Data management

According to Eq. (1), for data generation, we need to calculate weightage W_δ of individual VC/VP. Besides that, we have to calculate the individual value I_ζ and intensity of relative importance IRI_ζ of intensifying risk factors of every VC/VP. Before that, we must identify VC/VP from the engine subsystem. But, it is tough to choose every component of the engine as VC/VP because it is comprised of many components. In this case, we select the VC/VP as a subsystem that has outstanding significance to operate the vehicle smoothly based on the following parameters, i.e., the functionality, magnitude of losses, the magnitude of damages to the engine, and the magnitude of exposure to the users (Table 2).

To calculate the severity value for defining engine condition and maintenance priority, we need to calculate I_ζ and IRI_ζ utilising the intensifying factors of every VC/VP. We noticed that 13 intensifying or risk factors are closely related to the six VC/VP aforementioned. These could continue to contribute to fluctuating engine performance, as presented in Table 3.

For calculating I_ζ , the subject matter expert approach is a good option by collecting the opinions of the involved personnel in the automotive sector using the following questionnaire in Table 4. By following this approach, we employed MATLAB R2023b software tools based on random numbers to get I_ζ , whereas we set the probable value range considering influential factors impact if any subsystems fail (based on Tables 2 and 3). In this case, we generated 500 data of 13 intensifying factors of every VC/VP using the following questionnaire in Table 4 with a rating and indexing scale as None (0.0), Low (0.1–3.99), Medium (4.0–6.99), High (7.0–8.99), and Critical (9.0–10.0) from CSVS qualitative severity evaluation scale. On the other hand, before calculating IRI_ζ , RI_ζ of each VC/VP has been determined using the VFT method on a scale of 0 to 10. Then, we calculated IRI_ζ as

$$IRI_\zeta = \frac{RI_\zeta}{\sum_{\zeta=1}^{13} RI_\zeta} \quad (22)$$

where the $\sum_{\zeta=1}^{13} IRI_\zeta = 1$.

From these, the calculated average I_ζ and average RI_ζ for Cylinder and Piston (V_1) are presented in Table 5. A notable matter is that, though RI_ζ is calculated as the average of 500 data, but I_ζ value is presented based on the most frequent impact over the engine by following quantitative analysis of the qualitative data approach (Suter, 2012) to ensure consistency with the index rating.

Similarly, we calculated I_ζ , and RI_ζ for other VCs/VPs. Again, to calculate W_δ for individual VC/VP, 500 data for every item's relative

Table 2
Identification of VC/VP in engine subsystem.

Identified VC/VP	Probable issues	Identification framework			Reference
		Damage and loss	Potential problem	Functionality	
Cylinder & piston (V_1)	Its irregularities may raise oil consumption or influence overfueling. As a result may be inconsistent combustion, creating thermal and mechanical fatigue. Finally, these lead to excessive wear on pistons, cylinder, and piston rings, extreme piston noises, influenced other engine parts and reduced important part's reliability and longevity. Therefore, the engine required excessive maintenance expenses and affected overall engine performance.	Yes	Yes	Yes	Filipczyk and Stanik (2012)
Crankshaft (V_2)	It may create Excessive vibration, damaged bearings or essential engine parts, bent rotating parts, and mechanical fatigue failures. Therefore, these will lead to complex maintenance, and burden high maintenance costs.	Yes	Yes	Yes	Li et al. (2021)
Engine lubricant (V_3)	It may lead to engine performance degradation by excessive wear in mechanical components, corroded engine parts, seizing, welding or scuffing engine moving parts, reduced important parts reliability, damage to valuable parts, and engine useful service life, affecting the mechanical properties of other components, and so forth. Therefore, it leads to complex maintenance or sometimes needs to change the old engine with a new one.	Yes	Yes	Yes	Addison and Needelman (1986), ShashaVali, Patil, et al. (2021)
Engine overheating (V_4)	Due to overheating, the piston may seize, leading to thermostat failure, excessive and improper fuel consumption, damage to the valve train, engine detonations because of abnormal combustion, distort the head, affecting mechanical properties, and making it swell in the hottest areas. Also, blown head gaskets increase friction and wear, reduce spare parts' longevity and reliability, leading to complex maintenance and high maintenance costs.	Yes	Yes	Yes	Ebrinc and Cehreli (2007), Tan (2016)
Engine starter (V_5)	Due to the failure of the engine starter, unable to run the engine. Also, it may affect the battery life due to solenoid switch problems, vibration for improper mounting, and so forth.	Yes (Low)	Yes (Very Low)	Yes	Murugesan et al. (2012)
Engine misfires (V_6)	It may create abnormal combustion due to ignition faults and increase unburned hydrocarbons in the engine exhaust. Therefore, it decreased engine torque, created engine knocking, increased vibration and noise, decreased mileage, affected the environment, and reduced vehicle performance, reliability, and longevity of parts.	Yes	Yes	Yes	Filipczyk and Stanik (2012), Xu, Liu, Wu, Xing, and Wang (2019)

significance is generated among 0 to 10 scale using the VFT method and their average relative significance are presented in Table 6, where $\sum_{\delta=1}^6 W_{\delta} = 1$.

From Rahim et al. (2022), the degradation rate of VC/VP is 0.041 for every 10,000 km movement of a vehicle based on the fuel economy since the engine is assumed to operate for 30,000 km, so $k = 3$ and $\lambda = 0.041$. After putting the vulnerability value, VC/VP's degradation rate and k of the cylinder-piston issue in Eq. (1), the severity value for the cylinder-piston issue at t time is

$$SV_1(t) = S_1(t) * (1 - 0.041)^3 * 0.1599 * 6.48$$

$$SV_1(t) = 0.9141 * S_1(t) \quad (23)$$

In the same way, the severity value for crankshaft $SV_2 = 0.9341 * S_2$; for lubricants $SV_3 = 1.3824 * S_3$; for overheating $SV_4 = 1.5040 * S_4$; for starter $SV_5 = 0.5848 * S_5$; and for misfires issue $SV_6 = 0.4588 * S_6$ are calculated. Now we need the sensor reading from different VC/VPs which are described in the subsequent section.

4.1.2. Sensor reading accumulation

In the proposed VEHMS, sensor reading accumulation is an important task to detect engine health conditions and notify maintenance urgency. The sensors will remain activated during vehicular operation and collect data from designated points at specific time intervals for

further processing. In this context, the names of sensors and their purposes are presented in Table 7. These sensors will collect information and compare the real-time data with predefined VC/VP parameters. However, by considering these we generated more than 3000 sensor readings randomly using random function by MATLAB R2023b software tools. Furthermore, these generated data will be utilised in Eq. (1) to calculate the severity value, which is discussed in the subsequent section.

4.1.3. Sensor data utilisation

Before utilisation of the accumulated sensor data the notable matter is that the dimensions of the sensor data for different VC/VPs are not unique. Also, it would be very difficult to compare the state of individual VC/VPs to other VC/VPs due to their units' uniqueness. So, we labelled the sensor reading by 1,000 due to detecting the irregularities in the engine system for any VC/VP and 10 for good condition and consistent operation of the vehicle.

Now, we assumed the sensor detects irregularities at cylinder & pistons (V_1) and misfires (V_6) subsystem, where other VC/VP operates smoothly. Finally, the severity value of all VC/VPs at t time $SV_1 = 914.1$ ((crossed minimum value of critical threshold), $SV_2 = 9.341$, $SV_3 = 13.824$, $SV_4 = 15.040$, $SV_5 = 5.848$ and $SV_6 = 458.8$ (crossed the minimum threshold of moderate level but remains below critical threshold level)). Finally, the decision matrix for deciding about the vehicular engine's condition based on (Table 2) and the condition

Table 3
The details of VC/VPs' intensifying factors.

Intensifying factor name	Details for the selection of the intensifying factors
Cost Impact	The vehicular engine's vulnerable items' frequent purchasing or maintenance cost burdens the vehicle owner (Chatterton, Anable, Cairns, & Wilson, 2018).
Age Impact	The age of the vehicle, engine and running components significantly impacts engine performance because its performance inevitably deteriorates with the use time (Nixon & House, 1971). (How long the parts or components are used. The impact value will rise with increasing the duration of the use of vulnerable components.)
Damage Impact	Some components' irregularity might lead to intermittent, gradual, or sudden failure of other parts in the engine as a vulnerable component, whether it leads to any severe failure or not (Roy, Stark, Tracht, Takata, & Mori, 2016).
Reliability Impact	Reusing various old parts in the automobile could show inconsistent performance (Yafei, 2014). Whether the vulnerable component performing consistently or not? If good, the intensifying factor value may be the lowest; if bad, the value may rise. It could be measured based on the tenured history of vulnerable components.
Environmental Impact (Sound Pollution, Air Pollution, Water Pollution)	Whether the wastage of the engine components might harm the environment? Vehicle wheel dust (Enshassi, Kochendoerfer, & Rizq, 2014), lubricating oil (AY, Bates, & Kennedy, 1988), emission exhaust (Reşitoğlu, Altınışık, & Keskin, 2015), and many other wastes have a tremendous negative impact on the environment and human health.
Operational (Fresh or Reconditioned) Impact	Usually, new parts require less intensive maintenance than older ones. In the case of new spare parts, the impact value will be low, and the impact value will rise for the reconditioned parts.
Mechanical properties Impact	Does the irregularity of any engine components affect or influence the mechanical properties of other components (Ghassemieh, 2011)?
Risk Impact	Does the vulnerable component have any risk issues that may endanger the vehicle, its user, or pedestrians (Filtness et al., 2016; Papadimitriou et al., 2019; Pauer, Sipos, & Török, 2019)?
Safety Impact	Does the irregularity of any components deteriorate the safety of vehicle users and pedestrians (Wang, Jiang, Li, & Chen, 2018)?
Security Impact	Security is an essential consideration in the transport sector, so whether any components' irregularity breaches the vehicle, its users, and pedestrian security (Matsika, Ricci, Mortimer, Georgiev, & O'Neill, 2013)?
Comfortability Impact	Various issues like vibration, noise, thermal comfort (temperature), air quality, light, driving comfort, and many others affect the vehicle user's comfortability (Ormuž & Muftić, 2004; Rajhans, Rajhans, & Ahuja, 2009). Is the vulnerable component influencing comfort during operation?
Maintenance Complexity Impact	How complicated is maintenance due to the irregularity of vulnerable components? (Anthony & Cheung, 2017)
Spare Parts Availability	It is a critical issue to avoid any unexpected delay, so are the required components available in the local market, or do they need to be bought from overseas (Al-Bawi, 2015)?

mentioned in Eq. (1) is

$$\begin{bmatrix} 914.1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9.341 \\ 0 & 0 & 0 & 13.82 \\ 0 & 0 & 0 & 15.04 \\ 0 & 0 & 0 & 5.85 \\ 0 & 458.8 & 0 & 0 \end{bmatrix}$$

The notable matter that, $SV_{\delta}(t) \geq 745.3$, $745.3 < SV_{\delta}(t) \geq 407.79$, $407.79 < SV_{\delta}(t) \geq 99.0$ and $SV_{\delta}(t) < 99$ conditional limits utilised for defining the engine's respective VC/VP condition with critical problem, moderate problem, minor problem, or good condition, and plotted these in the above-stated matrix accordingly for t time. In this case, these values are assumed based on the minimum severity values of the most prominent VC/VP. On the other hand, according to Eq. (2), the sum of all elements of the matrix is $\sum VEHMS_D(t) = 1,416.95$, and it could define the real-time condition of the engine system.

4.2. Data labelling

In data labelling, we assumed the decision boundary threshold TH_D as the sum of all elements of the decision matrix. For example, $\sum VEHMS_D(t) \geq 925.29$, $925.29 < \sum VEHMS_D(t) \geq 478.79$, $478.79 < \sum VEHMS_D(t) \geq 99.0$ and $\sum VEHMS_D(t) \geq 99.0$ to define the condition of the vehicular engine, which may have critical, moderate, or minor problems (notify the stakeholders) or good conditions (no need to notify). In this case, these thresholds are assumed based on all VC/VP's functionality with consolidated minimum severity values. Since we achieved the severity value of the decision matrix's all element for t time is 1,416.95, which is valid for $\sum VEHMS_D(t) \geq 925.29$. Finally, it indicates that the vehicular engine faces a critical problem, though the misfire subsystem has a moderate problem. In this case, the proposed matrix will help us to identify which subsystem has a problem

and its severity. Similarly, we generated 3003 data by following the above procedure with t time interval.

4.3. Categorisations of engine health

To evaluate the model's performance, we labelled our vehicular engine condition data into four classes: good, minor, moderate, and critical. Our model can predict the exact engine condition based on the input features of the vehicular engine's physical condition. In our dataset, we have analysed and selected some features of the engine named "cylinder-piston", "crankshaft", "lubricants", "overheating", "starter", and "misfires." We modified each feature into text for further processing with our deep learning technique and passed it to the embedding layer. After processing the embedding and learning model, the final class of engine health is predicted. Later, we discussed the embedding, elaborately learning the model with aggregation and prediction layers.

4.4. Data visualisation

This section illustrates the visualisation of our data by focusing on the target class. The correlation matrix and bar chart visualisation are included here to understand the target data. Here, Fig. 3. shows the correlation among target classes of vehicles. For +1, a stronger positive relationship is indicated among target classes, whereas a stronger negative relationship is signified by -1. Another mentionable matter is that 0 has no relationship between those features (Schober, Boer, & Schwarte, 2018). Fig. 3 clearly shows that the correlation among different target classes lies between -0.38 to 0.37 out of -1 to 1, which indicates co-relation individually is moderate to weak, and this condition could help to enhance the decision accuracy of the proposed system.

Table 4
Questionnaire for Influential Factor's Individual Value (I_i).

Influential factors	Questionnaire	Rating	I_i	RI_i by VFT	IRI_i
Cost Impact	Is the vulnerable item expensive, or is the related maintenance expensive?				
Age Impact	Does the vulnerable item's age affect vehicle performance in addition to engine age? Like: For the new item, the considered value rating is 0. However, the performance is considered to deteriorate consecutively by 10,000 km intervals for none to a critical stage. For example, after 10,000 km, the rating will be low, medium for the next 10,000 km, high for after the next 10,000 km, and aggregately after 50,000 km considered critical. Please mention based on your experience for different mileage. 0 km 1st 10,000 km 2nd 10,000 km 3rd 10,000 km 4th 10,000 km 5th 10,000 km and more				
Damage Impact	Does the occurrence of irregularities have a negative damage effect on the vulnerable item?				
Reliability Impact	Does it make a difference whether the vulnerable item is new, used, or refurbished for the vehicle? Then, give the value of the negative impact that it had on the engine's performance; if it was new, give it a score of 0; if it was reconditioned or used, give a value score for reconditioned and used spare parts separately based on your experiences. new item used item reconditioned item				
Environmental Impact	Does the vulnerable item adversely impact the environment (if it is under operation or discarded as waste)?				
Operational Impact	Are the currently using vulnerable items fresh, reconditioned, or used? Then state the value of adverse impact on engine operational performance based on its conditions: if fresh, then 0; if reconditioned or used, provide value score for reconditioned used spare parts separately based on your experiences. new item used item reconditioned item				
Mechanical Impact	Do vulnerable item mechanical irregularities influence vehicle engine mechanical attributes or performance?				
Risk Impact	What risks does the vulnerable item pose to the vehicle, user, or pedestrian?				
Security Impact	Is the vulnerable item irregularity a security risk to the vehicle, user, or pedestrian?				
Comfortability Impact	Whether or not the vulnerable component affects the operation's comfort level.				
Maintenance Complexity Impact	How difficult is it to maintain vulnerable items due to their irregularity? In addition, does it have the power to cause complicated maintenance to fail by compromising other components?				
Spare Parts Availability	Is the vehicle brand vulnerable component available in the market? It is generally advisable to consider brand-specific when considering local and foreign brands. Local brand Foreign brand				

The number of target classes of vehicle engine health is presented in Fig. 4. In this figure, the X -axis shows the name of the desired class, while the Y -axis displays the total number of labelled classes.

4.5. GloVe embedding

For Deep CNN-BiGRU, the data must be embedded because the built data cannot directly accept the input. GloVe Vector is used for embedding the data. The GloVe Vector method is an unsupervised method in which the vector expresses the data. It illustrates the correlation of distance and semantic on useful space terms for each term (Pennington, Socher, & Manning, 2014). A matrix with the relevant term index is used to map each tokenised term from the embedding matrix. This embedding process has been done using GloVe Vector before mapping. Then, the converted vector from the GloVe vector is fed into the proposed learning model. The BiGRU layer is used as the first layer, and the

final layer is the hierarchical aggregation layer of the proposed system. The Deep CNN layer is placed in the middle of these two layers to build the hybrid deep learning model. Let a data term $S = w_1, w_2, w_3, \dots, w_N$ of the range N . The goal of this layer is to use d -dimensional vectors to describe every terms in S . Every term of each input data is taken as a static vector for embedding from the pre-trained embedded memory address. Then, the result of this embedded layer is the sequence of terms vectors $X = [x_1, x_2, x_3, \dots, x_N] \in R^{(N \times d)}$. The overall structure of our model is shown in Fig. 5. This figure demonstrates the organisation inside each framework, from the input phase through the prediction phase.

4.6. Deep CNN-BiGRU development

The suggested system includes an embedding layer with the GloVe vector. After the embedding layer, we employed the RNN layer for

Table 5
Intensifying factors I_{ζ} (Avg.), RI_{ζ} (Avg.) and IRI_{ζ} of cylinder-piston.

Intensifying factors for cylinder-piston	Most frequent impact ratings	Frequency	Individual value I_{ζ} (Avg.)	Relative importance RI_{ζ} (Avg.)	$IRI_{\zeta} = \frac{RI_{\zeta}}{\sum_{\zeta=1}^{13} RI_{\zeta}}$	$I_{\zeta} \times IRI_{\zeta}$
Cost Impact	Medium	367	6.03	7.6	0.09	0.55
Age Impact (3rd 10,000 km)	High	460	7.95	8.27	0.10	0.79
Damage Impact	High	489	7.98	8.4	0.10	0.80
Reliability Impact (Reused spare parts)	Critical	476	9.73	8.2	0.10	0.95
Environmental Impact	High	270	8.13	7.73	0.09	0.75
Operational Impact (Reused spare parts)	Critical	439	9.54	7.6	0.09	0.87
Mechanical properties Impact	Medium	288	5.13	5.93	0.07	0.36
Risk Impact	Low	468	2.37	4.8	0.06	0.14
Safety Impact	Low	433	2.06	4.6	0.06	0.11
Security Impact	No	495	0	1.73	0.02	–
Comfortability Impact	Medium	358	4.2	3.47	0.04	0.17
Maintenance Complexity Impact	High	369	8.06	7.93	0.09	0.76
Spare Parts Availability (Foreign brand)	Low	399	6.54	7.33	0.09	0.57
				83.59	1	6.84

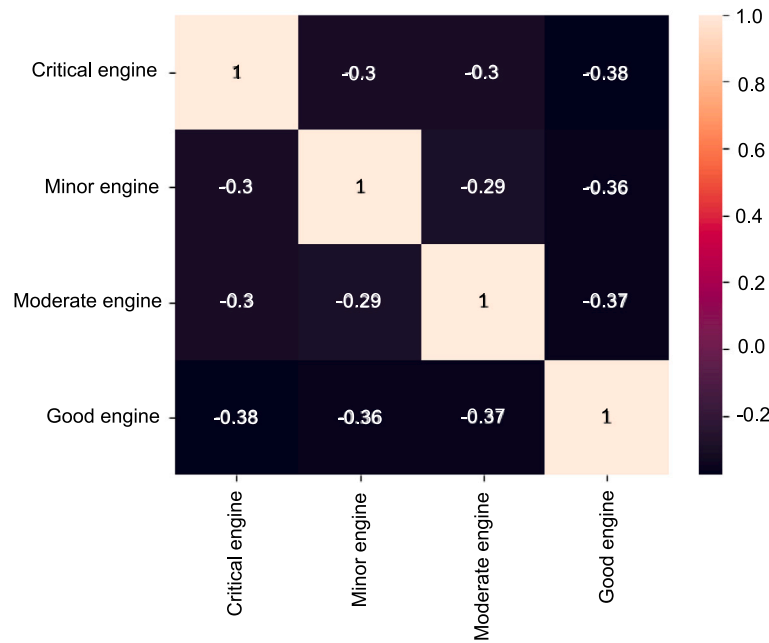


Fig. 3. Correlation among target classes of vehicle.

Table 6
Calculation of relative significance and W_{δ} of vehicular engine's VC/VP.

Engine's VC/VP	Relative significance (Avg.) by VFT	Weightage ($W_{\delta} = \frac{\text{Individual Relative Significance}}{\text{Total Relative Significance}}$)
Weightage (V_1)	6.82	0.1599
crankshaft (V_2)	7.63	0.1789
lubricants (V_3)	9.38	0.2199
overheating (V_4)	9.44	0.2213
starter (V_5)	4.60	0.1078
misfires (V_6)	4.79	0.1123
Total=	42.66	$\sum_{\delta=1}^6 W_{\delta} = 1$

BiGRU. Initially, we selected BiGRU with 64 units for the first layer and 128 units for the next layer. Then, the Deep CNN layer is concatenated with the layer with a scale of 64 units. When cell and batch size are 128 in this framework, we employ four epochs for classification tasks. The notable matter is that, Fig. 2 shows the total operating sequences of our suggested new Deep CNN-BiGRU model for the training, validation, and testing phases.

5. Experimental setup and result analysis

5.1. Experimental setup

This research is mainly designed to evaluate the decision performance of the developed model by employing the Deep CNN-BiGRU approach as the hybrid deep learning technique. To analyse the obtained results and strengthen our argument, we employed some advanced deep learning and hybrid deep learning, i.e., CNN, LSTM, BiLSTM, GRU, BiGRU, CNN-LSTM, and Attention CNN-LSTM, and their experiment setup is described as follows through the profound scrutinisation.

5.1.1. Deep learning approach

In the Big Data era, the deep learning approach is the prominent option to monitor and classify the engine health condition utilising the generated massive amount of structural data. This approach is based on the multi-stage neural networks as filtering and classification phases. This filtering phase usually extracts information from the fed dataset through the convolution and pooling layers. On the other hand, a few fully connected (FC) layers utilised the MLP in the classification phase

Table 7
Selected Sensors for VEHMS Data Accumulation.

Name of VC/VP	Sensor name	Reading purposes	References
cylinder-piston (V_1)	Accelerometer sensors	Perform wavelet transform analysis to measure signal's peak value over time, RMS, mean, Kurtosis, and Skewness	Moosavian et al. (2016)
crankshaft (V_2)	Accelerometer sensors	Perform wavelet transform analysis to measure signal's peak value over time, RMS, mean, Kurtosis, and Skewness	Moosavian et al. (2016)
lubricants (V_3)	Online lubricants quality monitoring sensors	Physio-chemical parameters of lubricants	Perić, Nedić, and Grkić (2014)
overheating (V_4)	Voltage variation, coolant level and coolant temperature sensors	Voltage at the terminal of the cooling fan and water circulation pumps, coolant volume, and its temperature	Bayir and Omer (2011), Castiglione, Pizzonia, and Bova (2016), Fusiek, Niewczas, and Judd (2012), Hossain, Smith, and Davies (2017)
starter (V_5)	Voltage variation sensor	Voltage at the terminal	Bayir and Omer (2011)
misfires (V_6)	Magnetic pickup sensor	Mean output power techniques for a dynamic engine	Boudaghi, Shahbakhti, and Jazayeri (2015), Song, Liu, and Yan (2017)

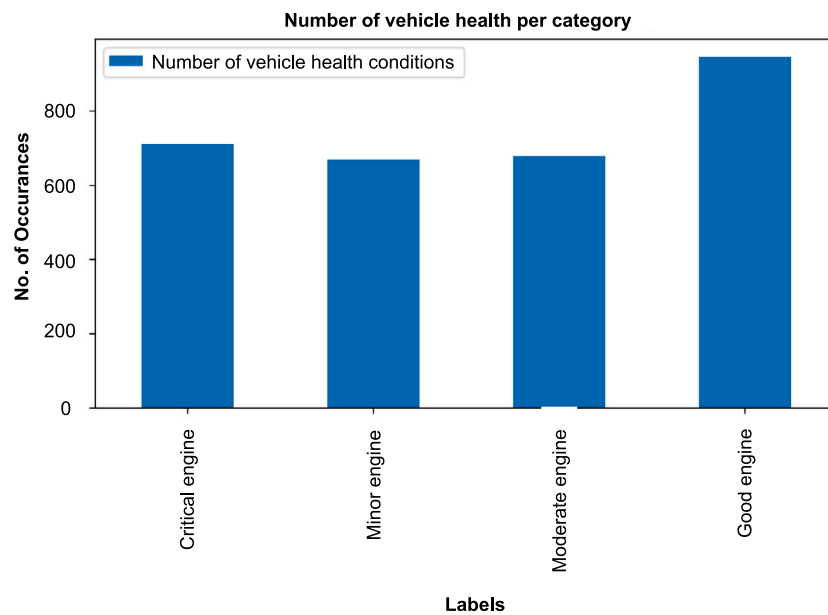


Fig. 4. Bar chart visualisation to show the number of target classes of vehicle.

to categorise the engine condition based on the fed data. In this case, the ReLU activation function transforms the logit value nonlinearly at the convolution layer. The softmax activation function transforms logits of the neurons at the output layer for multiclass categorisation of engine health by matching with the probability distribution. 0.8:0.1:0.1 of the total data was fed to the system for training, validation, and testing purposes with 50 epochs to profoundly scrutinise the VEHMS decision model's accuracy utilising the CNN, LSTM, BiLSTM, GRU and BiGRU approach.

5.1.2. Hybrid deep learning approach

Since we planned to employ several hybrid deep learning approaches, the experimental setup for all approaches cannot be described here due to space complexities. So, the experimental setup of Deep CNN-BiGRU here is the prominent approach to reveal the performed evaluation process to the respected reader. The layer sequences are included in this approach to enhance and improve the decision outcomes. So, initially, the input data is embedded as the sequence with the index value of the matrix using the GloVe vector for compressing the space of the input feature into smaller ones. In this case, the notable matter is that the total data is divided into 0.8:0.1:0.1 for feeding this model as training, validating data, and testing respectively, besides

other approaches mentioned above and evaluating their performances. All letters are converted to lowercase in the preprocessing stage, and the input sequence is then converted to fixed-size sequences. Moreover, the dropout layers (Johnson & Zhang, 2016) are used to regularise the input text sequence.

Then, it is dropped out and sent to the Bidirectional GRU layer. The RNN's BiGRU operation output is transmitted to the deep CNN layer. The Deep CNN's convoluted output is sent into a hierarchical aggregation layer. Finally, the output is transmitted to the next layer for prediction using a dense layer with a softmax activation function. In our multiclass problem, the cross-entropy loss function was used for predicting class labels with an ADAM optimiser to minimise the loss. The learning rate was 0.01. Besides, the ROC evaluation was performed to analyse the skilfulness of the decision model for the Deep CNN-BiGRU approach. Model hyperparameters are presented in Table 8 with layer type, output shape, and the total parameters of the proposed model (Deep CNN-BiGRU). Model hyperparameters are presented in Table 8 with presenting layer type, output shape, and the total parameters of the proposed model (Deep CNN-BiGRU). Batch size, epoch, data split, activation function, optimisers and loss function area are the same for all the models implemented. The model consists of a total of 2,633,304 parameters, which occupy 10.05 MB of memory. Out

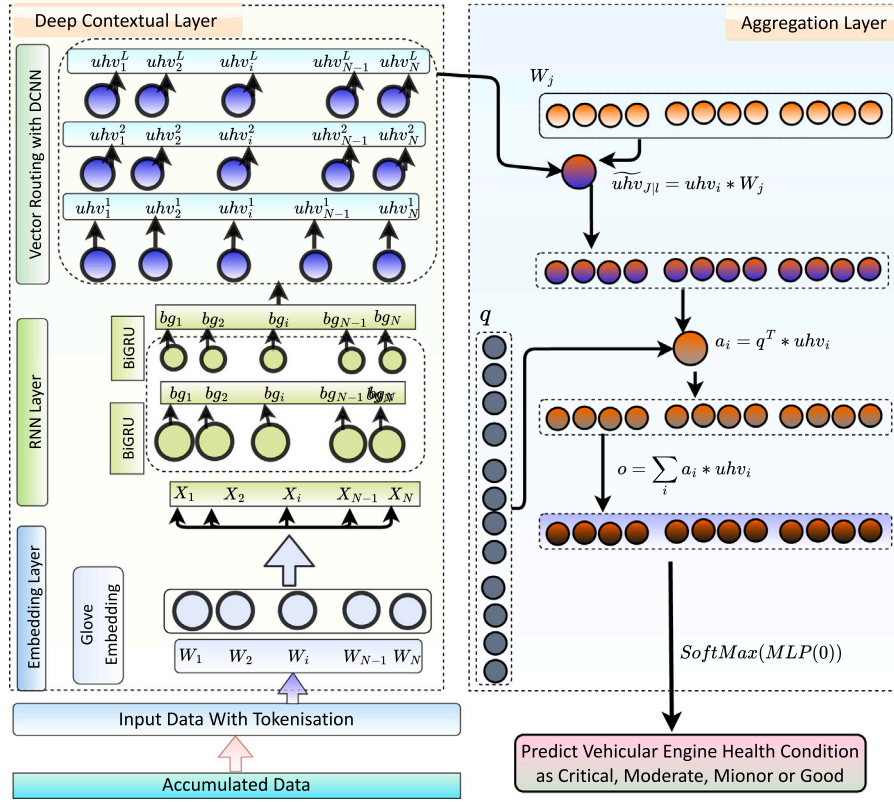


Fig. 5. The architecture of the Deep CNN-BiGRU approach.

Table 8

Model hyper parameter.

Layer type	Output shape	Parameters
embedding_2 (Embedding)	(None, 200, 128)	2560000
spatial_dropout1d_2 (Spatial Dropout1D)	(None, 200, 128)	0
conv1d_2 (Conv1D)	(None, 200, 100)	51300
bidirectional_2 (Bidirectional GRU)	(None, 200, 50)	19050
batch_normalization_2 (Batch Normalisation)	(None, 200, 50)	200
global_max_pooling1d_2 (GlobalMaxPooling1D)	(None, 50)	0
dropout_2 (Dropout)	(None, 50)	0
dense_4 (Dense)	(None, 50)	2550
dense_5 (Dense)	(None, 4)	204

of these, 2,633,204 parameters are trainable, accounting for 10.04 MB. The remaining 100 parameters are non-trainable, with a size of 400 bytes. The model is trained with a batch size of 32 over 50 epochs.

5.2. Results analysis

5.2.1. Result analysis using deep learning approaches

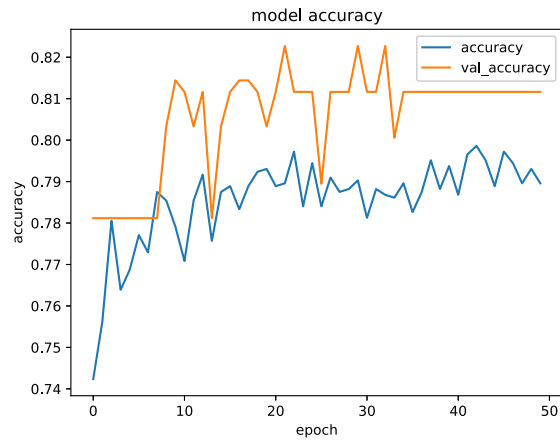
For CNN, the obtained decision accuracy, and losses for 50 epochs of training and validation cases are presented in Fig. 6. The blue line marks the outcomes of the decision model for the training phase, and the orange line marks the validation phase. Though Fig. 6(a) initially shows that the decision accuracy for both training and validation phases was very low, after subtle epochs, these were increased outstandingly and provided higher decision accuracy with nearly 0.80

until the last epoch during the validation phases than training phases (almost 0.79).

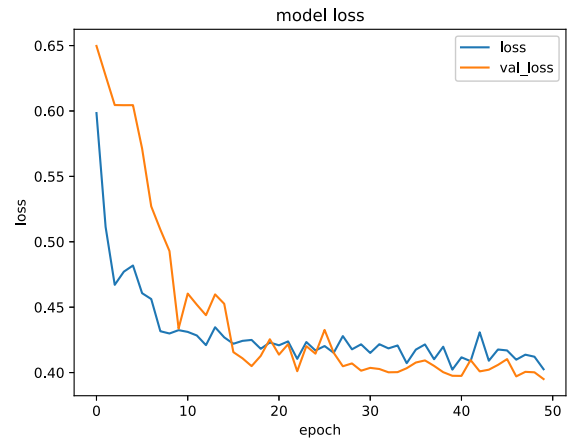
Again, Fig. 6(b) shows the decision loss using CNN, where both curves for the training and validation phases decreased significantly. The satisfactory matter is that the decision loss for the validation phase was lower than the training phase from 15th epoch. The overall decision accuracy at the training phase is 0.7815, and the loss is 0.4209. Still, Fig. 6(b) shows downgrading the loss curves instead of a stable condition, which indicates this approach has significant inconsistency for a good fit. Therefore, we employed other approaches to achieve a better outcome for this research, such as LSTM and BiLSTM.

The LSTM and BiLSTM are considered to mitigate the gradient problem by exploiting backpropagation over time. This approach used memory blocks in the layer instead of neurons and helped straightforwardly diagnose vehicular engine irregularities. Moreover, the experimental setup for these two approaches is followed like CNN and the data is fed with the same ratio. Figs. 7 and 8 show the results obtained as decision accuracy and loss from the LSTM and BiLSTM techniques, respectively. When comparing Fig. 7(a) with Fig. 8(a), for the case of decision accuracy, the movement of the validation line over 50 epochs is a minor but noticeable variation. We observed that the LSTM model achieved an accuracy of 0.8471, while the BiLSTM model attained a lower accuracy of 0.7988. Similarly, in terms of loss performance, the LSTM model recorded a loss of 0.3887, whereas the BiLSTM model exhibited a higher loss of 0.4150. An opposite coincidence has appeared for the decision loss of LSTM and BiLSTM, presented in Fig. 7(b) and Fig. 8(b). Though the gap between the training and validation phases in these figures is very small, the loss is significant. For that, we tried further investigating by employing GRU and BiGRU approaches.

Following the same experimental setup for the individual deep learning approaches, we feed the data to GRU and BiGRU systems. In this case, We observed that the GRU model achieved an accuracy of 0.8050, while the BiGRU model attained a slightly higher accuracy of 0.8390. Similarly, in terms of loss performance, the GRU model

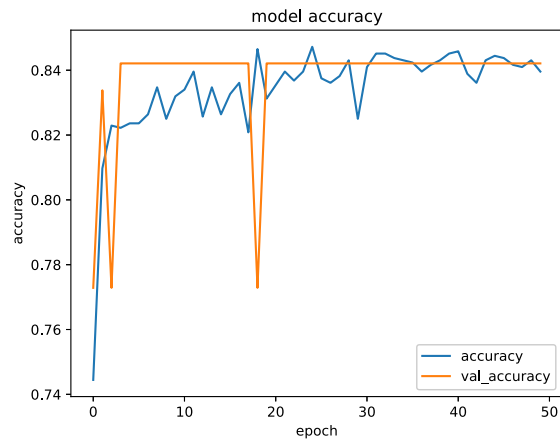


(a) Decision Accuracy

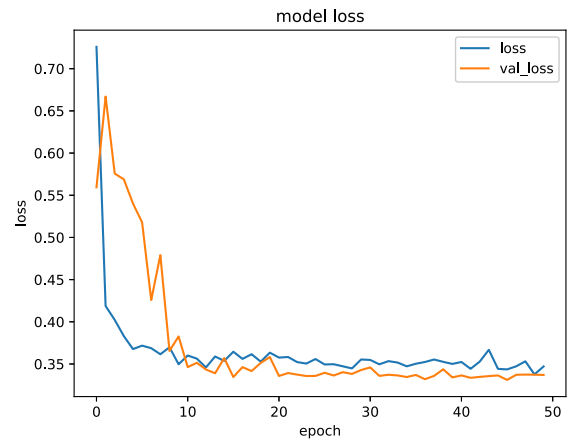


(b) Decision Loss

Fig. 6. Decision outcome of the developed model using CNN.

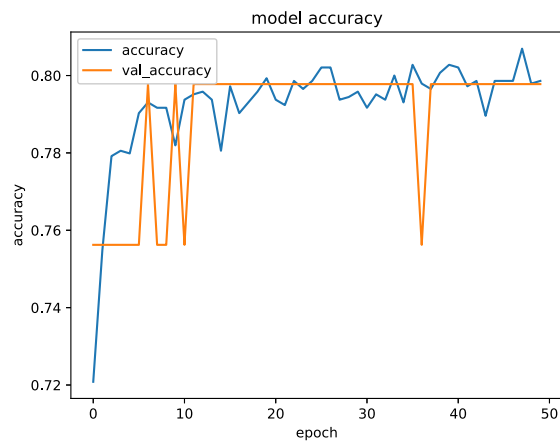


(a) Decision Accuracy

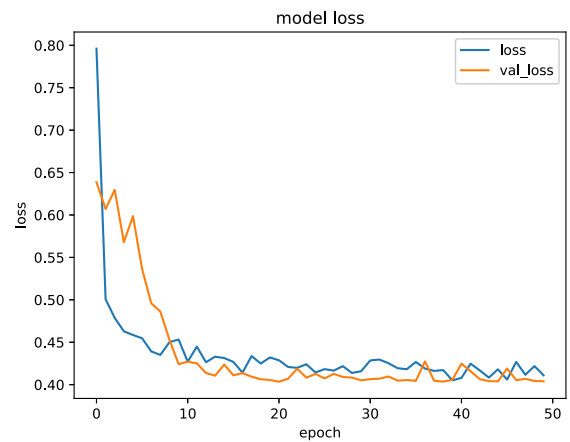


(b) Decision Loss

Fig. 7. Decision outcome of the developed model using LSTM.



(a) Decision Accuracy



(b) Decision Loss

Fig. 8. Decision outcome of the developed model using BiLSTM.

recorded a loss of 0.3754, whereas the BiGRU model exhibited a lower loss of 0.3129. To scrutinise the decision outcome over the 50 epochs, we presented both decision accuracy and decision loss curves in Figs. 9 and 10 for GRU and BiGRU, respectively.

Again, the graphical presentation of decision loss for GRU shows that this approach is underfitting due to the continuous downgrading of the training phase. On the other hand, from Fig. 10, the BiGRU showed a good fit for the VEHMS decision model by providing stable decision

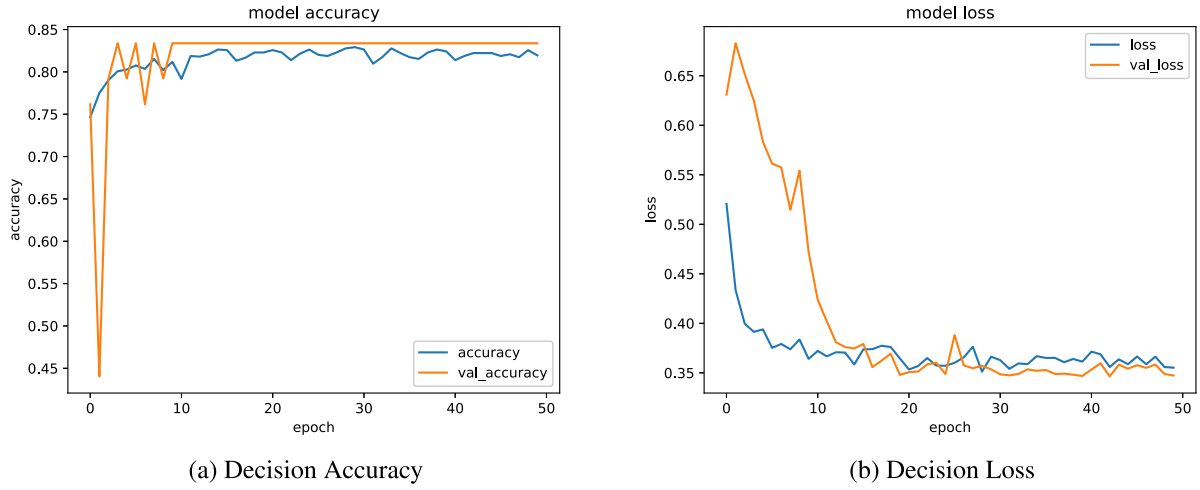


Fig. 9. Decision outcome of the developed model using GRU.

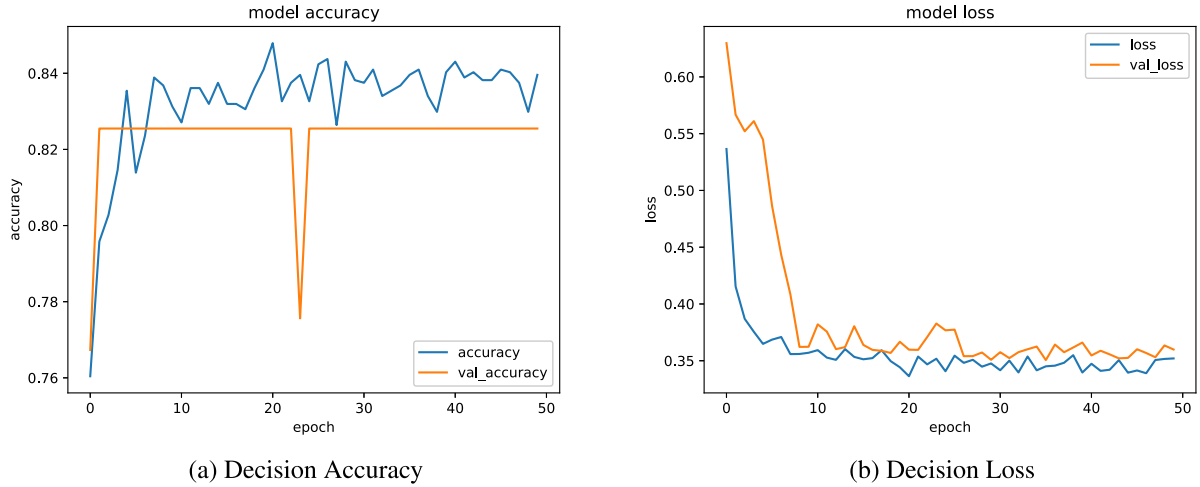


Fig. 10. Decision outcome of the developed model using BiGRU.

accuracy and losses over 50 epochs. Though the generalisation gap is very small for both techniques, the loss seems substantial. In this case, we planned to use the hybrid deep learning approach to mitigate these problems by increasing decision accuracy and minimising losses.

5.2.2. Result analysis using hybrid deep learning approaches

In CNN-LSTM techniques, the data is sent straight to the LSTM layer of 64 cells, and then the output of that layer is sent to the CNN layer of the same size. At the end of the validation phase, the engine condition is classified into four categories using the *softmax* activation function and cross-entropy loss function using the ADAM optimiser, with a decision accuracy of 0.8055 and a decision loss of 0.4181. The decision accuracy and losses of the CNN-LSTM approach are presented in Fig. 11 for 50 epochs to scrutinise further investigation. In Fig. 11(b), the gap between the training and validation loss curve is growing from epoch 15 to 50. Also, these loss curves are continuously downgrading.

To mitigate such issues and improve decision accuracy, we introduced an extra layer as the attention layer (Xie et al., 2021). In this case, the decision accuracy with decision losses for 50 epochs is presented in Fig. 12 to be understood clearly. Though a significant transition appeared for the decision accuracy of the validation phase at the 25th epoch, it stabilised within 0.8563, during the training phase it got 0.8352. Some fluctuations appeared in the decision loss. In the training phases, it continued to downgrade instead of stability, indicating that this is an overfitting condition. Lastly, we want to

validate the result by employing the Deep CNN-BiGRU approach in the subsequent paragraph.

Now, as per the stated experimental setup of Deep CNN-BiGRU, we fed the data to the system and obtained outstanding decision accuracy and decision losses for both the training and testing phases, and these are presented in Fig. 13. Through the profound scrutinisation of Fig. 13(a) for the decision accuracy of the VEHMS decision model, we noticed that the decision accuracy for the training phase fluctuated between approximately 0.90 and 0.88. Nonetheless, the validation phase's maximum and minimum decision accuracy were almost 0.905 and 0.8750, respectively. However, there is a lower minimum in the training phase than in the validation phase. That means the decision accuracy for the validation phase improved compared to the training phases, except for a few epochs among 50 epochs. That indicates the decision model with Deep CNN-BiGRU generalised finely. Conversely, Fig. 13(b) indicates that the decision loss during the validation phase is minor than during the training phase. The difference between validation loss and training loss is relatively small and overlaps the training phase loss from 15th epoch to until the end. The discussed matters infer that the Deep CNN-BiGRU approach perfectly fits the VEHMS decision model to categorise the engine health condition based on the structural data. However, based on the discussed parameters and decision accuracy, this technique outperformed the previously discussed approaches. Our method obtained the very lowest 0.2618 loss during training.

Again, to scrutinise the Deep CNN-BiGRU outcomes for individual classes, we presented the decision accuracy with decision losses of

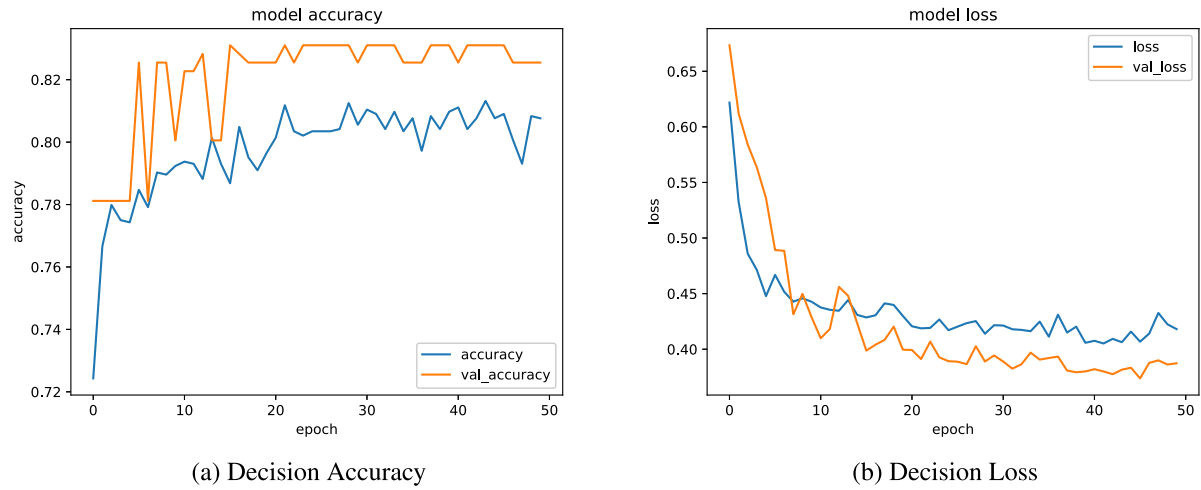


Fig. 11. Decision outcome of the developed model using CNN-LSTM.

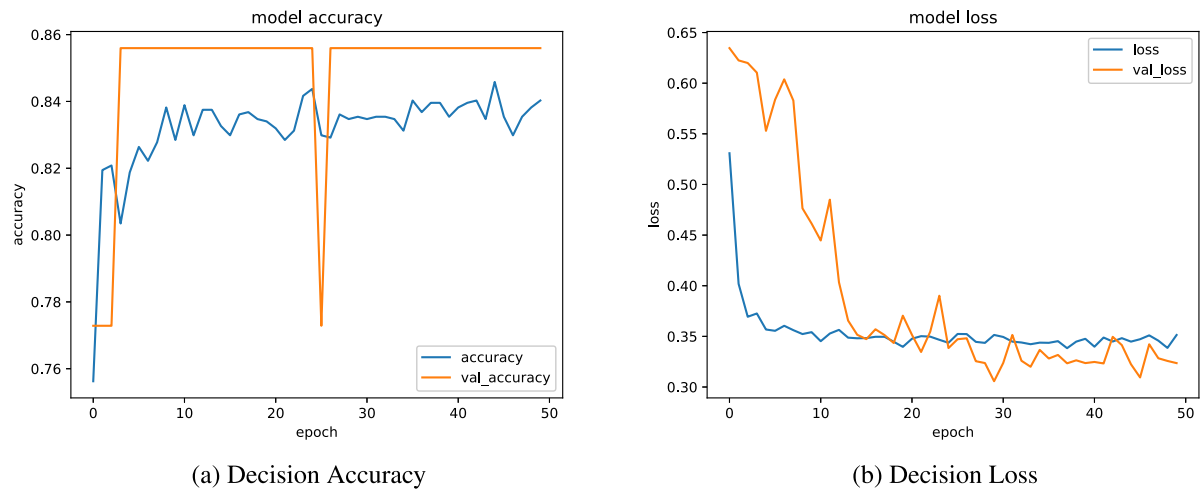


Fig. 12. Decision outcome of the developed model using attention CNN-LSTM.

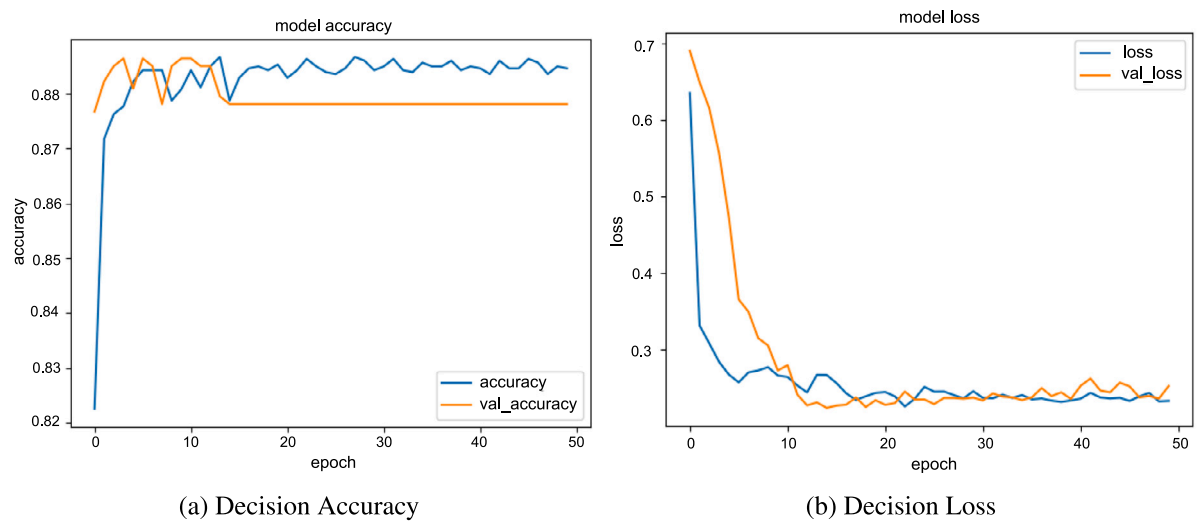
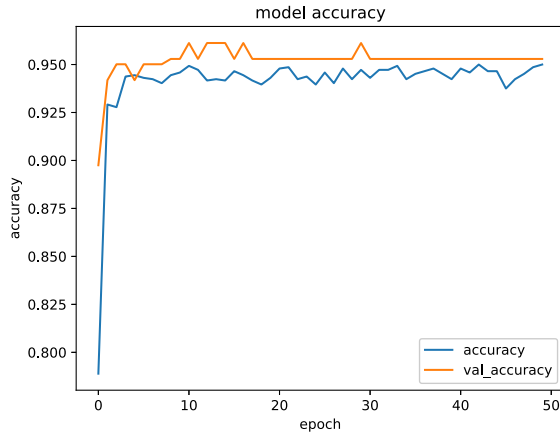


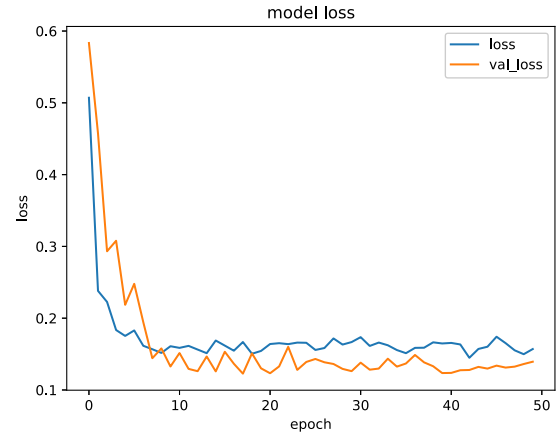
Fig. 13. Decision outcome of the developed model using Deep CNN-BiGRU.

individual classes for vehicular engine multiclass conditions. Generally, sigmoid activation is used for binary class, but we employed it at the output layer with binary-cross entropy optimiser instead of softmax

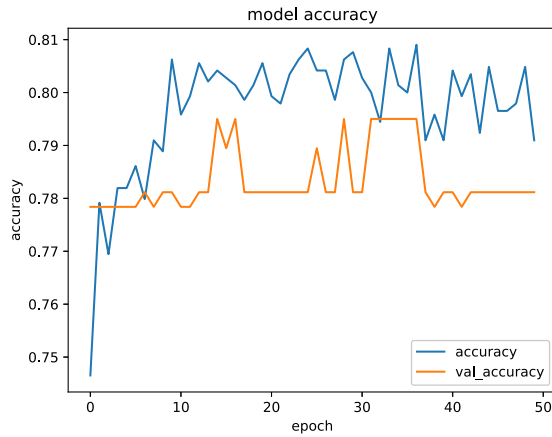
activation function and cross-entropy loss function with ADAM optimiser. In this case, each binary target class is trained separately, and those separate classifications of the different classes are integrated for



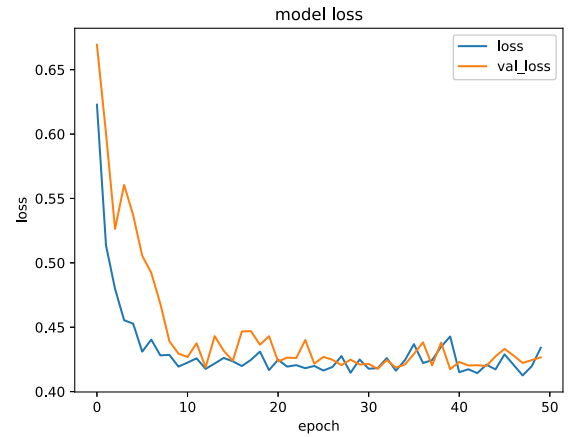
(a) Model accuracy for critical class



(b) Model loss for critical class



(c) Model accuracy for moderate class



(d) Model loss for moderate class

Fig. 14. Decision accuracy and decision losses for individual classes (critical and moderate).

multiclass output. We used each class categorisation separately for our proposed multiclass and presented the decision accuracy and losses with Deep CNN-BiGRU in Figs. 14 and 15. Ultimately, the decision accuracy, decision losses, mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), coefficient of determination (R^2), precision, recall, and F1 score for the individual vehicular engine health condition predictions are presented in Table 9.

5.3. Discussion

In this research, we selected CNN, LSTM, BiLSTM, GRU, BiGRU, CNN-LSTM, Attention CNN-LSTM, and Deep CNN-BiGRU approaches as the prominent techniques of deep learning algorithms to analyse the performance of the developed VEHMS decision model. Table 10 presents the achieved decision accuracy and loss, advantages, as well as disadvantages of these employed approaches for the testing phase.

After profoundly scrutinising these results, we found the lowest decision accuracy is 0.7815 with the highest decision loss of 0.4209 for CNN. Also, from the previous discussion, it is evident that this approach is not feasible for employing a vehicular engine health condition monitoring system due to its underfitting conditions. The third-highest decision accuracy is almost 0.8471, achieved by LSTM approaches. Additionally, the second-highest decision accuracy is almost 0.8563, achieved by attention CNN-LSTM approaches. Nevertheless, we achieved outstanding performance from the Deep CNN-BiGRU approach provided a maximum decision accuracy of 0.8897 to categorise

the vehicular engine health condition with a minimum decision loss of 0.2618. Also, this approach is a good fit and is well-performed based on the fundamental requirements for monitoring the vehicular engine efficiently. Again, the generated ROC curve is presented in Fig. 16, and the obtained micro-average area is 0.97 and the macro-average is 0.96 out of 1. That indicates the decision model achieved outstanding skilfulness with Deep CNN-BiGRU to monitor the vehicular engine health condition in real-time and categorise its condition as the engine has good, minor, moderate or critical conditions.

6. Conclusion

This article introduces novel research to align with Industry 4.0 standards in the automotive sector, employing a hybrid deep learning algorithm for real-time vehicular engine fault diagnosis. In this case, the novelty of this work is that we considered associated risk factors, degradation rates of vulnerable engine components, and real-time sensor readings to develop the VEHMS decision model. Based on that the contribution of this research is

- proposed an efficient VEHMS decision model for monitoring vehicular engine health using a hybrid deep learning technique.
- established an enhanced theoretical background to achieve satisfactory decision accuracy with minimal losses.
- presented an organised methodology to analyse data behaviour and obtain satisfactory results.

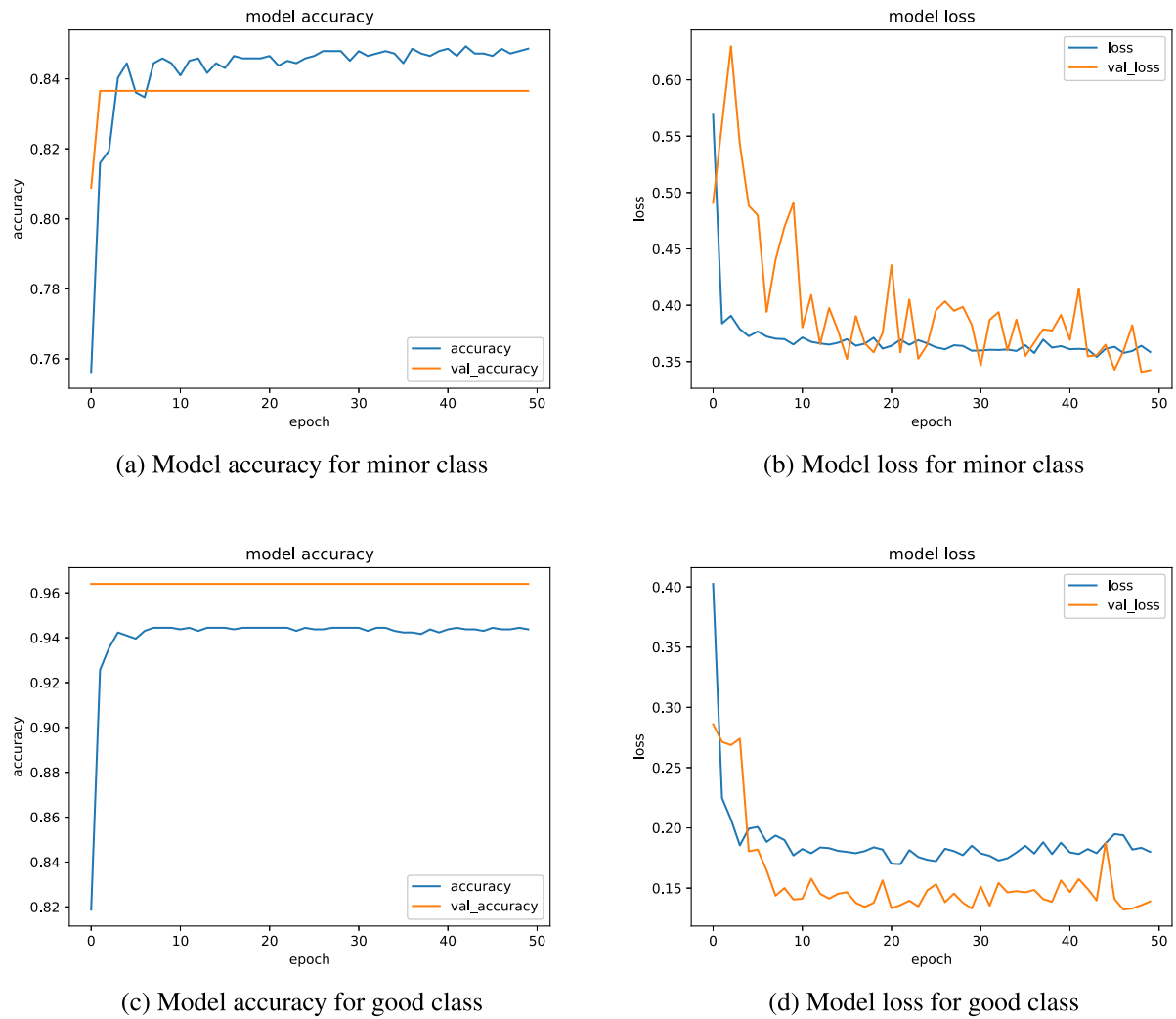


Fig. 15. Decision accuracy and decision losses for individual classes (minor and good).

Table 9

Performance of Deep CNN-BiGRU for Individual Target Classes.

Training performance									
Class	Accuracy	Loss	MAE	MSE	RMSE	R2	Precision	Recall	F1 Score
Critical	0.9494	0.1413	0.0816	0.0363	0.1906	0.7966	0.9500	0.9500	0.9500
Minor	0.8598	0.3565	0.2433	0.1084	0.3293	0.3862	0.8700	0.8500	0.8500
Moderate	0.8039	0.4146	0.2813	0.1354	0.3679	0.2062	0.7800	0.8000	0.7700
Good	0.9483	0.1673	0.0837	0.0450	0.2123	0.7909	0.9500	0.9500	0.9500
Overall	0.8956	0.3117	0.1934	0.0812	0.2750	0.5449	0.8875	0.8875	0.8800
Validation performance									
Class	Accuracy	Loss	MAE	MSE	RMSE	R2	Precision	Recall	F1 Score
Critical	0.9384	0.1322	0.0840	0.0354	0.1882	0.7965	0.9500	0.9400	0.9400
Minor	0.8269	0.3683	0.2524	0.1143	0.3881	0.3401	0.8700	0.8300	0.8400
Moderate	0.7853	0.4368	0.3017	0.1441	0.3796	0.2163	0.7900	0.7600	0.7500
Good	0.9450	0.2006	0.0880	0.0486	0.2204	0.7788	0.9500	0.9500	0.9500
Overall	0.8739	0.2844	0.1815	0.0856	0.2940	0.5329	0.8900	0.8700	0.8700
Testing performance									
Class	Accuracy	Loss	MAE	MSE	RMSE	R2	Precision	Recall	F1 Score
Critical	0.9617	0.1148	0.0741	0.0288	0.1699	0.8490	0.9600	0.9600	0.9600
Minor	0.8352	0.3541	0.2428	0.1086	0.3296	0.3284	0.8800	0.8400	0.8500
Moderate	0.8103	0.4157	0.2843	0.1354	0.3680	0.2340	0.8000	0.8100	0.7800
Good	0.9517	0.1627	0.0797	0.0424	0.2620	0.8000	0.9500	0.9500	0.9500
Overall	0.8897	0.2618	0.1702	0.0788	0.2823	0.5863	0.8975	0.8900	0.8850

- selected an optimal technique to accurately categorise vehicular engine health and provide precise decisions to minimise maintenance expenses.

As a result, the proposed Deep CNN-BiGRU algorithm-based VEHMS decision model demonstrated impressive decision accuracy of 0.8897 for 80 per cent training data, suggesting that this approach could

Table 10
Comparative analysis of models.

Model	Accuracy	Loss	Advantage	Disadvantage	Overview
CNN	0.7815	0.4209	Minimal time complexity when autonomously grasp hierarchical features directly.	Limited ability to capture sequential dependencies.	Leverages convolutional layers for feature extraction from input data, followed by fully connected layers.
LSTM	0.8471	0.3887	Capture long-term dependencies in sequential data by retaining and selectively forgetting information over time.	Susceptibility to vanishing gradient problems for long ranged feature.	Excels in classifying numeric data by capturing temporal patterns and long-term dependencies effectively.
BiLSTM	0.7988	0.4150	Effectively capturing temporal dependencies and contextual information, leading to improved accuracy.	Higher computational requirements and slower training times.	Captures bidirectional context, considering past and future information simultaneously.
GRU	0.8050	0.3754	Providing simpler architecture and faster training times while still effectively capturing temporal dependencies.	Potentially limiting its ability to capture complex temporal patterns.	Efficiently handles numeric data classification tasks by gating mechanisms and simplified memory management.
BiGRU	0.8390	0.3129	Providing enhanced modelling of temporal dependencies for improved accuracy and faster convergence.	Vanishing gradient issues in deeper networks can hinder its effectiveness in capturing long-range dependencies.	Adeptly handles numeric data classification tasks with improved memory efficiency and context awareness.
CNN-LSTM	0.8055	0.4181	Automatically extract relevant features from input sequences.	Have the risk of overfitting with large parameters and cannot capture long term dependencies.	CNN-LSTM hybrid model leverages CNN for feature extraction and LSTM for sequence modelling.
Attention CNN-LSTM	0.8563	0.3650	Enable superior feature extraction and deftly capture extensive temporal relationships.	May involve higher computational demands and increased model complexity	Integrates attention mechanism with CNN for feature extraction and LSTM for sequence modelling.
Deep CNN-BiGRU	0.8897	0.2618	Facilitate robust representation learning and capture intricate patterns in the data.	May include increased computational complexity and potential overfitting.	Deep CNN-BiGRU combines Deep CNNs for feature extraction with BiGRU for sequence modelling, effectively classifying numeric data with context awareness.

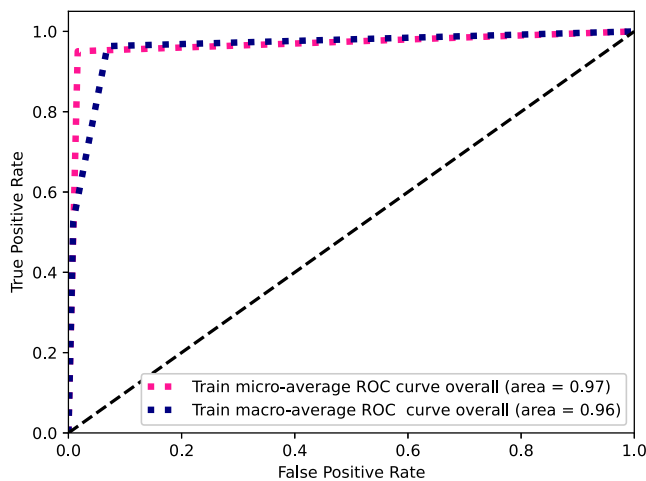


Fig. 16. ROC-AUC curve of Deep CNN-BiGRU for VEHMS Decision Model.

effectively diagnose engine health conditions and notify stakeholders based on the urgency of engine conditions for maintenance.

Beyond these, there are significant opportunities for implementing the Deep CNN-BiGRU algorithm-based structural health monitoring system in autonomous vehicles and railway systems, managing big data, and advancing the automotive sector. This research has limitations, including its focus solely on the engine subsystem with only six features. There is considerable scope for future work, such as examining complete vehicular systems with additional features and incorporating a more extensive array of sensors. By addressing these limitations, further advancements can be made in vehicle health monitoring, ultimately

improving the reliability and efficiency of transportation systems. Additionally, there is scope to identify more influential factors beyond the current 13 and to explore alternative approaches, rather than relying solely on the subject matter expert approach, to obtain more accurate data.

CRedit authorship contribution statement

Md. Abdur Rahim: Conception of the project, Model development, Data preparation, Analysis, Writing – original draft. **Md. Mustafizur Rahman:** Analysis, Writing – review & editing, Funding acquisition, Supervision, validation. **Md. Shofiqul Islam:** Coding, Investigation, Validation, Writing – original draft. **Abu Jafar Md. Muzahid:** Visualization, Writing – review & editing. **Md. Arafatur Rahman:** Conception of the project, design of methodology, Analysis, Writing – review & editing, Resources, Validation. **Devarajan Ramasamy:** Supervision, Funding acquisition, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: MD MUSTAFIZUR RAHMAN reports financial support was provided by University Malaysia Pahang Al-Sultan Abdullah. MD MUSTAFIZUR RAHMAN reports a relationship with the University Malaysia Pahang Al-Sultan Abdullah that includes employment and funding grants. MD MUSTAFIZUR RAHMAN has patent N/A pending to N/A. The authors express their sincere gratitude to the University of Malaysia Pahang Al-Sultan Abdullah for their laboratory support and funding assistance provided through the University Distinguished Research Grants (Project number RDU223016) as well as UK-Saudi Challenge Fund 2024.

Data availability

Data will be made available on request.

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